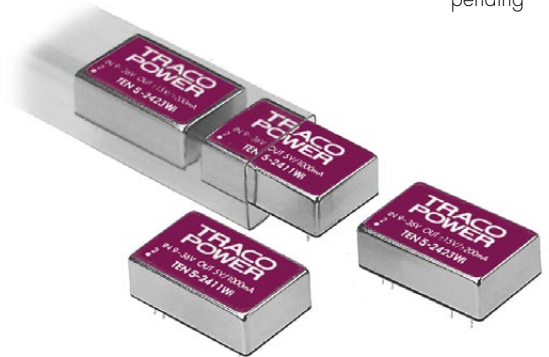


Features

- ◆ Ultra wide 4:1 Input Range
- ◆ Full SMD-Design
- ◆ High Efficiency up to 84%
- ◆ Extended Operating Temperature Range -40°C to +85°C
- ◆ Indefinite Short-circuit Protection
- ◆ I/O-Isolation 1'500 VDC
- ◆ Input Filter meets EN 55022, Class A and FCC, Level A without external Components
- ◆ Shielded Metal Case with insulated Baseplate
- ◆ 24-pin DIP with Industry Standard Pinout
- ◆ Lead free Design, RoHS compliant
- ◆ 3 Year Product Warranty



The TEN 5WI series is a family of dc-dc converter modules with 5 to 6 W output power featuring ultra wide 4:1 input voltage ranges. They come in a compact DIL-24 metal package with industry-standard footprint. A high efficiency up to 84% allows operating temperatures from -40°C to 85°C at full output power without derating. A built-in EMI input filter complies with EN 55022, class A without need of external components. Further standard features include over voltage protection and continuous short circuit protection. Typical applications for these converters are battery operated equipment and distributed power architectures in communication and industrial electronics, everywhere where isolated, tightly regulated voltages are required.

Models

Ordercode	Input voltage range	Output voltage	Output current max.	Efficiency typ.
TEN 5-2410WI	9 – 36 VDC	3.3 VDC	1200 mA	76 %
TEN 5-2411WI		5 VDC	1000 mA	79 %
TEN 5-2412WI		12 VDC	500 mA	84 %
TEN 5-2413WI		15 VDC	400 mA	83 %
TEN 5-2421WI		± 5 VDC	± 500 mA	79 %
TEN 5-2422WI		± 12 VDC	± 250 mA	84 %
TEN 5-2423WI		± 15 VDC	± 200 mA	83 %
TEN 5-4810WI	18 – 75 VDC	3.3 VDC	1200 mA	76 %
TEN 5-4811WI		5 VDC	1000 mA	79 %
TEN 5-4812WI		12 VDC	500 mA	84 %
TEN 5-4813WI		15 VDC	400 mA	83 %
TEN 5-4821WI		± 5 VDC	± 500 mA	79 %
TEN 5-4822WI		± 12 VDC	± 250 mA	84 %
TEN 5-4823WI		± 15 VDC	± 200 mA	83 %

Input Specifications

Input current no load /full load	24 Vin models: 20 mA typ. 48 Vin models: 10 mA typ.
Start-up voltage / under voltage shut down	24 Vin models: 9 VDC / 8.5 VDC typ. 48 Vin models: 18 VDC / 16 VDC typ.
Surge voltage (1 sec. max.)	24 Vin models: 50 V max. 48 Vin models: 100 V max.
Reverse voltage protection	1.0 A max.
Conducted noise (input)	EN 55022 level A, FCC part 15, level A

Output Specifications

Voltage set accuracy	± 2.0 % max.
Regulation	– Input variation Vin min. to Vin max. ± 0.3 % max. – Load variation 10 – 100 % single output models: ± 2.0 % max. dual output models: ± 2.0 % max. balanced load ± 3.0 % max. unbalanced load
Ripple and noise (20 MHz Bandwidth)	75 mVpk-pk max
Temperature coefficient	± 0.02 % / K
Current limitation	> 110% of Iout max., constant current
Short circuit protection	indefinite (automatic recovery)
Capacitive load	3.3 /5 VDC models 1000 µF max. 12 /15 VDC models 470 µF max. ±5 VDC models 220 µF max. (for each output) ±12 /±15 VDC models 100 µF max. (for each output)

General Specifications

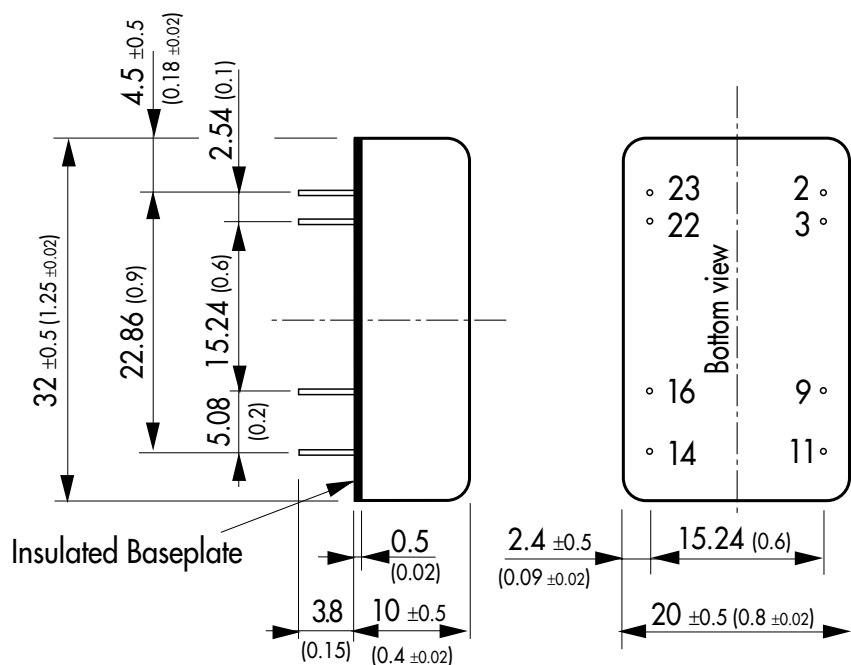
Temperature ranges	– Operating – 40 °C ... + 85 °C – Case temperature + 100 °C max. – Storage – 55 °C ... + 125 °C
Derating	3.5% /K above 70°C
Humidity (non condensing)	95 % rel H max.
Reliability, calculated MTBF (MIL-HDBK-217 E)	>1 Mio. h @ + 25 °C
Isolation voltage	Input/Output 1'500 VDC
Isolation capacity	Input/Output 350 pF typ
Isolation resistance	Input/Output (500 VDC) > 1'000 M Ohm
Switching frequency	300 kHz typ. (Pulse frequency modulation PFM)
Safety standards	UL/cUL 60950 , IEC 60950, EN 60950

All specifications valid at nominal input voltage, full load and +25°C after warm-up time unless otherwise stated.

Physical Specifications

Case material	Steel, Nickel plated
Baseplate material	non conductive FR4
Potting material	Epoxy (UL 94V-0 rated)
Weight	17 g (0.49 oz)
Soldering temperature	max. 260 °C / 10 sec.

Outline Dimensions mm (inches)



Pin-Out		
Pin	Single	Dual
2	-Vin (GND)	-Vin (GND)
3	-Vin (GND)	-Vin (GND)
9	No pin	Common
11	No function	-Vout
14	+Vout	+Vout
16	-Vout	Common
22	+Vin (Vcc)	+Vin (Vcc)
23	+Vin (Vcc)	+Vin (Vcc)

Pin diameter $\varnothing 0.5 \pm 0.05$ (0.02 ±0.002)

Tolerances ± 0.5 (0.02)

Specifications can be changed without notice